



ECEN 314 - Signals and Systems

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Module 2 - System Analysis Using Laplace Transforms

Learning Objectives



- Be able to find the Laplace Transform of common elementary functions.
- Be able to find the inverse Laplace transform of rational polynomial functions of s , using partial fractions, properties of LT's, and table look-up.
- Be able to use various properties to help find the Laplace Transform of more complicated signals.
- Be able to use Laplace techniques to find the zero-state (forced) response of any system whose input/output relationship is described by a differential equation.

Laplace Transforms



It is common to use Laplace Transforms to aid in the analysis of circuits/systems. Recall for a signal $x(t)$ defined on $0 \leq t < \infty$, its Laplace Transform is defined as

$$\mathcal{L}\{x(t)\} = \int_0^{\infty} x(t) e^{-st} dt.$$

Examples:

$$\begin{aligned} \text{➤ } x(t) = u(t) \text{ (step function)} &= \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases} \leftrightarrow X(s) = \int_0^{\infty} e^{-st} dt = \left. \frac{e^{-st}}{-s} \right|_0^{\infty} = \frac{1}{s} \\ &\text{for } \operatorname{Re}[s] > 0, \end{aligned}$$

$$\begin{aligned} \text{➤ } x(t) = e^{-at} u(t) &\leftrightarrow X(s) = \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = \left. \frac{e^{-(s+a)t}}{-(s+a)} \right|_0^{\infty} = \frac{1}{(s+a)} \\ &\text{for } \operatorname{Re}[s] > -a, \end{aligned}$$

$$\begin{aligned} \text{➤ } x(t) = \cos(\omega t) u(t) &\leftrightarrow X(s) = \int_0^{\infty} \cos(\omega t) e^{-st} dt = \frac{e^{-st}}{s^2 + \omega^2} (\omega \sin(\omega t) - s \cos(\omega t)) \Big|_0^{\infty} = \frac{s}{s^2 + \omega^2}, \\ &\text{for } \operatorname{Re}[s] > 0, \end{aligned}$$

$$\begin{aligned} \text{➤ } x(t) = \sin(\omega t) u(t) &\leftrightarrow X(s) = \int_0^{\infty} \sin(\omega t) e^{-st} dt = \frac{e^{-st}}{s^2 + \omega^2} (-s \sin(\omega t) - \omega \cos(\omega t)) \Big|_0^{\infty} = \frac{\omega}{s^2 + \omega^2}, \\ &\text{for } \operatorname{Re}[s] > 0, \end{aligned}$$

These and a few other relevant Laplace Transforms are summarized in the tables on the next slides.

Table of Common Laplace Transforms



Signal, $x(t)$	Transform, $X(s)$	Region of Convergence
$u(t)$ – unit step function	$\frac{1}{s}$	$\text{Re}[s] > 0$
$\delta(t)$ – unit delta function	1	All s
$t^n u(t), \quad n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$	$\text{Re}[s] > 0$
$t^{n-\frac{1}{2}} u(t), \quad n = 1, 2, 3, \dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)\sqrt{\pi}}{2^n s^{n+1/2}}$	$\text{Re}[s] > 0$
$e^{at} u(t)$	$\frac{1}{s-a}$	$\text{Re}[s] > a$
$e^{j\omega t} u(t)$	$\frac{1}{s-j\omega}$	$\text{Re}[s] > 0$
$t^n e^{at} u(t), \quad n = 1, 2, 3, \dots$	$\frac{n!}{(s-a)^{n+1}}$	$\text{Re}[s] > a$
$t^n e^{j\omega t} u(t), \quad n = 1, 2, 3, \dots$	$\frac{n!}{(s-j\omega)^{n+1}}$	$\text{Re}[s] > 0$

Table of Common Laplace Transforms



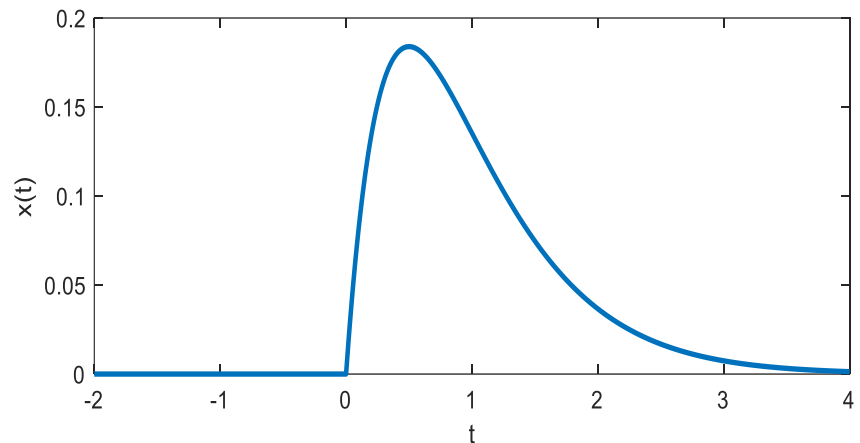
Signal, $x(t)$	Transform, $X(s)$	Region of Convergence
$\cos(\omega t) u(t)$	$\frac{s}{s^2 + \omega^2}$	$\text{Re}[s] > 0$
$\sin(\omega t) u(t)$	$\frac{\omega}{s^2 + \omega^2}$	$\text{Re}[s] > 0$
$\cosh(bt) u(t)$	$\frac{s}{s^2 - b^2}$	$\text{Re}[s] > b $
$\sinh(bt) u(t)$	$\frac{\omega}{s^2 - b^2}$	$\text{Re}[s] > b $
$e^{at} \cos(\omega t) u(t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$	$\text{Re}[s] > a$
$e^{at} \sin(\omega t) u(t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$	$\text{Re}[s] > a$
$e^{at} \cosh(bt) u(t)$	$\frac{s - a}{(s - a)^2 - b^2}$	$\text{Re}[s] > a + b $
$e^{at} \sinh(bt) u(t)$	$\frac{\omega}{(s - a)^2 - b^2}$	$\text{Re}[s] > a + b $

Laplace Transforms

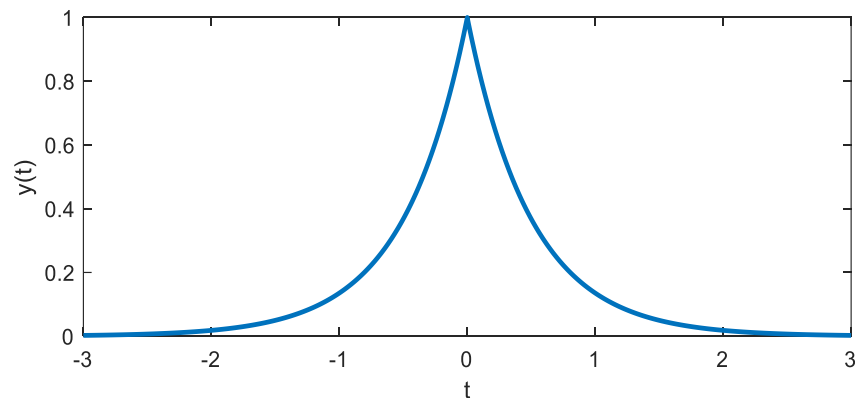


- Notice that all of the time domain signals in the table have unit step functions attached to them.
- This is to emphasize the one-sided nature of the Laplace Transform.
- Since the Laplace Transform was defined as an integral from 0 to ∞ , it only applies to signals that are 0 for $t < 0$ (right-sided).
- Signals that are left-sided or two-sided do not have a Laplace Transform. We will later introduce a different transform to deal with those types of signals.

A one-sided signal: $x(t) = te^{-2t}u(t)$



A two-sided signal: $y(t) = e^{-2|t|}$



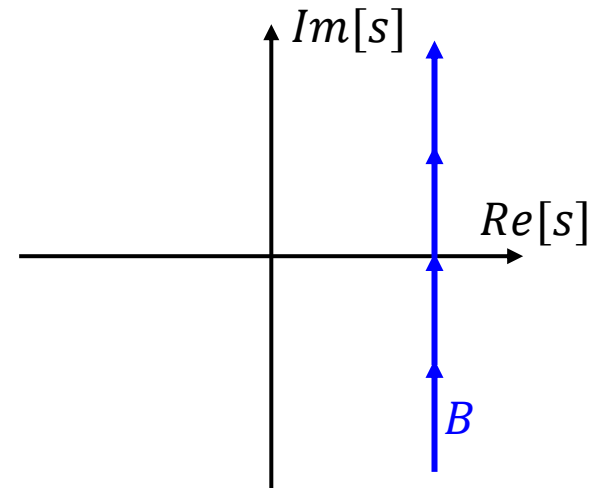
Inverse Laplace Transforms



The inverse Laplace transform is written in terms of a complex contour integral in the s -plane

$$\mathcal{L}^{-1}\{X(s)\} = \frac{1}{2\pi j} \oint_B X(s) e^{st} ds.$$

The contour of integration is the so-called Bromwich contour which is a vertical line in the complex plane located anywhere to the right of all of the singular points of $X(s)$.



We almost never use the contour integral to compute inverse Laplace transforms. Rather, we manipulate the given transform $X(s)$ into known forms (table entries) and then use a table lookup to compute inverse transforms.

Partial fractions are used extensively to help manipulate functions of s which are in the common form of rational fractions (quotients of polynomials involving s).

Example

$$\begin{aligned} X(s) &= \frac{1}{s^2 + 2s + 5} \\ &= \frac{1}{(s + 1)^2 + 4} \\ &= \frac{1}{2} \cdot \frac{2}{(s + 1)^2 + 2^2} \\ x(t) &= \frac{1}{2} e^{-t} \sin(2t) u(t) \end{aligned}$$

Quick Review of Partial Fractions



- For rational functions of s (where the order of the numerator is smaller than the order of the denominator) with no repeated roots in the denominator, the function can be expressed in the form

$$F(s) = \frac{N(s)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

- In such cases, the function can be expressed via partial fractions in the form

$$F(s) = \frac{A_1}{s-p_1} + \frac{A_2}{s-p_2} + \cdots \frac{A_n}{s-p_n} \quad \text{where } A_k = \lim_{s \rightarrow p_k} (s - p_k)F(s).$$

Example 2.1: Find the partial fraction expansion of $F(s) = \frac{s+3}{s^2+3s+2}$

Answer:

$$F(s) = \frac{2}{s+1} - \frac{1}{s+2}.$$

Quick Review of Partial Fractions



- If denominator has repeated roots, then there is a similar but more complicated result

$$F(s) = \frac{N(s)}{(s - p_1)^{r_1} (s - p_2)^{r_2} \cdots (s - p_n)^{r_n}}$$

- In this case, the function is expressed via partial fractions in the form

$$F(s) = \sum_{k=1}^n \left\{ \frac{A_{k,1}}{s - p_k} + \frac{A_{k,2}}{(s - p_k)^2} + \cdots \frac{A_{k,r_k}}{(s - p_k)^{r_k}} \right\}$$

$$\text{where } A_{k,m} = \frac{1}{(r_k - m)!} \lim_{s \rightarrow p_k} \frac{d^{r_k - m}}{ds^{r_k - m}} (s - p_k)^{r_k} F(s).$$

Example 2.2: Find the partial fraction expansion of $F(s) = \frac{s}{s^2 + 4s + 4}$

Answer:

$$F(s) = \frac{1}{s+2} - \frac{2}{(s+2)^2}.$$

Inverse Laplace Transforms



Example 2.3 - Find the inverse Laplace Transforms for

a) $X(s) = \frac{s}{s^2 + 4s + 3},$

b) $Y(s) = \frac{s}{s^3 + s^2 + s + 1},$

Answers:

a) $x(t) = \left[\frac{3}{2}e^{-3t} - \frac{1}{2}e^{-t} \right] u(t),$

b) $y(t) = \frac{1}{2}[\cos(t) + \sin(t) - e^{-t}]u(t).$

Properties of Laplace Transforms



For more complicated signals that do not appear in tables of Laplace Transforms, various properties are used to write the complicated signals in terms of more basic signals.

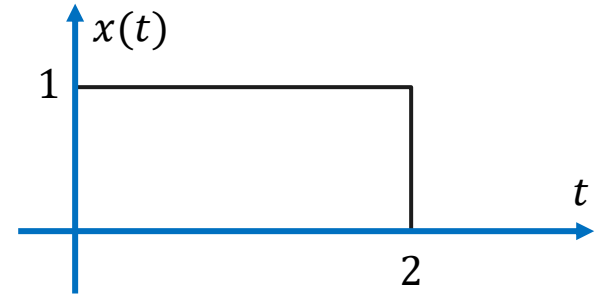
Properties of Laplace Transforms are then combined with the previous tables to get the transform of more complicated signals.

Property	Signal, $x(t)$	Transform, $X(s)$
Linearity	$ax(t) + by(t)$	$aX(s) + bY(s)$
Time Shift	$x(t - t_o)$	$X(s)e^{-st_o}$
Frequency Shift	$x(t)e^{s_o t}$	$X(s - s_o)$
Time Scaling	$x(\alpha t), \alpha > 0$	$\frac{1}{\alpha} X\left(\frac{s}{\alpha}\right)$
Differentiation	$\frac{dx}{dt}$	$sX(s) - x(0^-)$
Integration	$\int_0^t x(\tau) d\tau$	$\frac{X(s)}{s}$
Multiplication by t	$tx(t)$	$-\frac{dX}{ds}$
Modulation	$x(t) \cos(\omega_o t)$ $x(t) \sin(\omega_o t)$	$\frac{1}{2} [X(s - j\omega_o) + X(s + j\omega_o)]$ $\frac{j}{2} [X(s + j\omega_o) - X(s - j\omega_o)]$
Convolution	$x(t) * y(t)$	$X(s)Y(s)$
Multiplication	$x(t)y(t)$	$X(s) * Y(s)$

Laplace Transforms



Example 2.4 - Find the Laplace Transform for the rectangular signal shown.

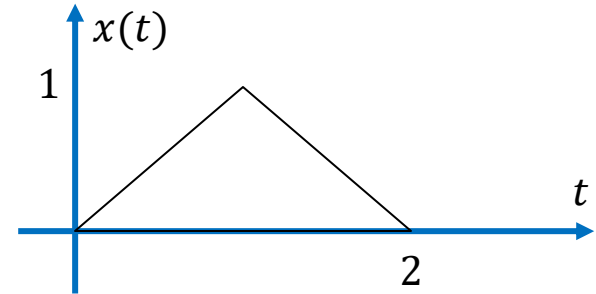


Answer: $X(s) = \frac{1}{s}(1 - e^{-2s})$

Laplace Transforms



Example 2.5 - Find the Laplace Transform for the triangular signal shown.



Answer: $X(s) = \left(\frac{1-e^{-s}}{s}\right)^2$

Laplace Transforms



Example 2.6 - Find the Laplace Transform for
 $x(t) = t\cos(2\pi t)u(t)$.

Answer: $X(s) = \frac{s^2 - (2\pi)^2}{(s^2 + (2\pi)^2)^2}$

System Analysis Using Laplace Transforms



A system is something that takes an input, $x(t)$ and produces some output, $y(t)$.



We often work with systems whose input/output relationships are described by a differential equation (circuits are great examples of this).

$$b_n \frac{d^n y}{dt^n} + \dots + b_2 \frac{d^2 y}{dt^2} + b_1 \frac{dy}{dt} + b_0 y(t) = a_m \frac{d^m x}{dt^m} + \dots + a_2 \frac{d^2 x}{dt^2} + a_1 \frac{dx}{dt} + a_0 x(t)$$

Taking L.T. of both sides results in

$$\begin{aligned} Y(s) \sum_{k=0}^n b_k s^k - y(0) \sum_{k=1}^n b_k s^{k-1} - \frac{dy}{dt}(0) \sum_{k=2}^n b_k s^{k-2} - \dots - \frac{d^{n-1}y}{dt^{n-1}}(0) b_n \\ = X(s) \sum_{k=0}^m a_k s^k - x(0) \sum_{k=1}^m a_k s^{k-1} - \frac{dx}{dt}(0) \sum_{k=2}^m a_k s^{k-2} - \dots - \frac{d^{m-1}x}{dt^{m-1}}(0) a_m. \end{aligned}$$

Define the polynomials $b(s) = \sum_{k=0}^n b_k s^k$ and $a(s) = \sum_{k=0}^m a_k s^k$. Then

$$Y(s) = X(s) \frac{a(s)}{b(s)} + \text{terms that depend on I.C.'s.}$$

System Analysis Using Laplace Transforms



$$Y(s) = X(s) \frac{a(s)}{b(s)} + \text{terms that depend on I.C.'s.}$$

There are two parts of the output:

- ✓ Zero-State Response (Forced response) = when all ICs are zero, due only to input, $x(t)$.
- ✓ Zero-Input Response (Natural Response) = when input is zero, due only to ICs.

We are often most interested in the response of a circuit/system to a given input. So, moving forward, we will assume all ICs are 0 and focus on the Zero-State/Forced Response

$$Y(s) = X(s)H(s) \quad \text{where } H(s) = \frac{a(s)}{b(s)}.$$

The term $H(s) = \frac{a(s)}{b(s)}$ is what we call the *transfer function* of the system.

System Analysis Using Laplace Transforms



Example 2.7: Find the forced response (zero-state), $y(t)$, of the system described by

$$y''(t) + 2y'(t) + 2y(t) = x(t),$$

when the input is $x(t) = e^{-t}u(t)$.

Answer: $y(t) = [1 - \cos(t)]e^{-t}u(t)$

Bonus Material: Circuit Analysis Using Laplace Transforms



An important property of Laplace Transforms is that it transforms derivatives into multiplication by s as seen below:

$$\begin{aligned}\mathcal{L}\left\{\frac{dx}{dt}\right\} &= \int_0^{\infty} \frac{dx}{dt} e^{-st} dt \\ &= x(t)e^{-st} \Big|_0^{\infty} + s \int_0^{\infty} x(t)e^{-st} dt && \text{(using integration by parts once).} \\ &= -x(0) + sX(s).\end{aligned}$$

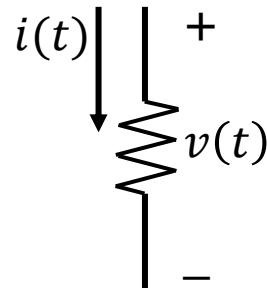
In a similar way we can also show that integration corresponds to dividing by s in the Laplace domain.

In the following, we will use this property of Laplace transforms to observe the behavior of typical circuit elements in the Laplace domain. The first, resistors, is trivial:

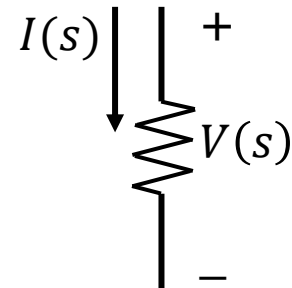
Resistors:

Time domain: $v(t) = Ri(t)$

Laplace domain: $V(s) = RI(s)$



Time Domain



Laplace Domain

Bonus Material: Circuit Analysis Using Laplace Transforms



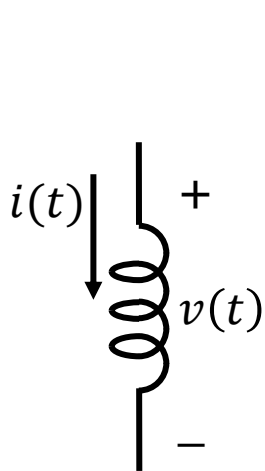
Inductors:

Time domain: $v(t) = L \frac{di}{dt}$

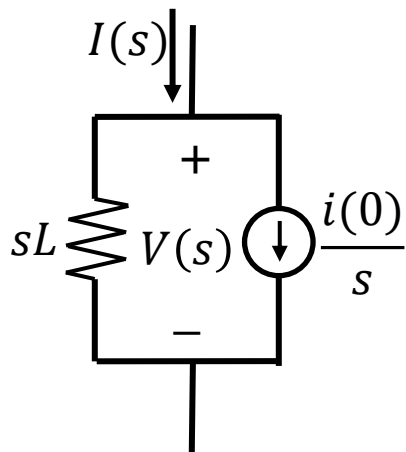
Laplace domain:

$$V(s) = sLI(s) - Li(0)$$

$$I(s) = \frac{V(s)}{sL} + \frac{i(0)}{s}$$



Time Domain



Laplace Domain

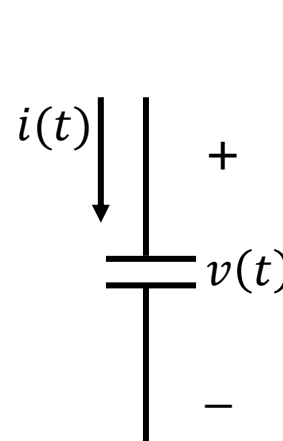
Capacitors:

Time domain: $i(t) = C \frac{dv}{dt}$

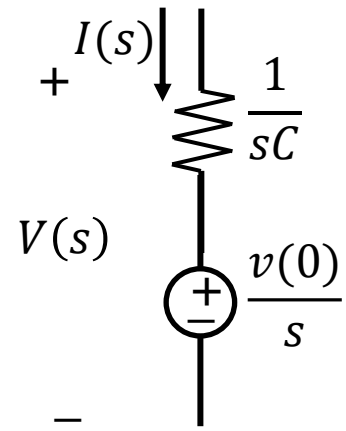
Laplace domain:

$$I(s) = sCV(s) - Cv(0)$$

$$V(s) = \frac{I(s)}{sC} + \frac{v(0)}{s}$$



Time Domain

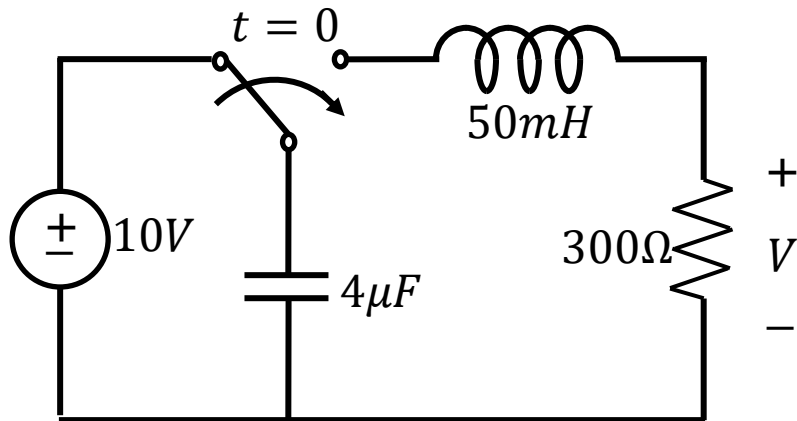


Laplace Domain

Bonus Material: Circuit Analysis Using Laplace Transforms

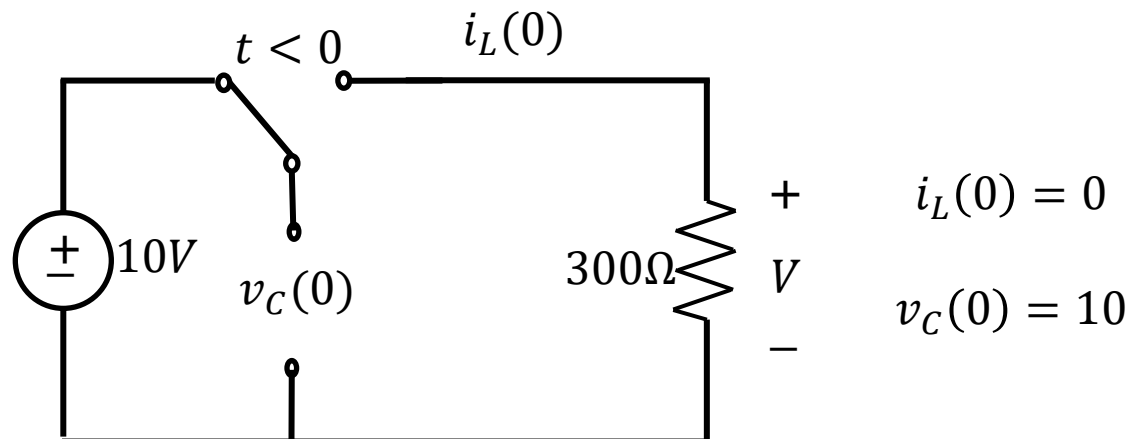


Here is an example of a 2nd order circuit that we can analyze using Laplace domain techniques.



In the circuit shown, steady state has been reached before the switch moves at time $t = 0$. Find the voltage on the resistor for $t > 0$

Step 1: Find the initial inductor current and capacitor voltage using steady state techniques



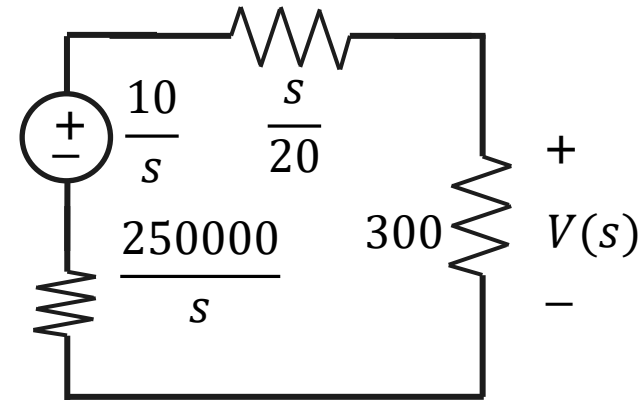
$$i_L(0) = 0$$
$$v_C(0) = 10$$

Bonus Material: Circuit Analysis Using Laplace Transforms



Step 2: Re-draw the circuit in the Laplace domain for $t > 0$.

Note: In this case there is no current source in parallel with the inductor since we found $i_L(0) = 0$.



Step 3: Solve for the voltage/current of interest using resistive circuit techniques.

$$V(s) = \frac{10}{s} \cdot \frac{300}{\frac{s}{20} + 300 + \frac{250000}{s}} \quad (\text{using a voltage divider})$$

$$= \frac{60000}{s^2 + 6000s + 5000000} \quad (\text{simplifying})$$

$$= \frac{60000}{(s + 5000)(s + 1000)} \quad (\text{factoring})$$

Bonus Material: Circuit Analysis Using Laplace Transforms



Step 4: Bring voltage/current of interest back to the time domain by finding the inverse Laplace transform.

$$V(s) = \frac{60000}{(s + 5000)(s + 1000)}$$
$$= \frac{15}{s + 1000} - \frac{15}{s + 5000}$$

(the previous step was accomplished using partial fractions)

$$v(t) = [15e^{-1000t} - 15e^{-5000t}]u(t)$$

- Since we found the roots of the denominator (poles) to be real, we are expecting exponential solutions in the time domain.

- As such, we should manipulate the form of $V(s)$ to match the form of an exponential in our table of Laplace Transforms.

- The relevant entry in the table is:

$$e^{-at} \leftrightarrow \frac{1}{s+a}.$$

➤ Using the Laplace approach allows us to find both the transient and the steady state components (or the forced and the natural components) of the voltage. If we used phasor techniques, we only get the steady state response.

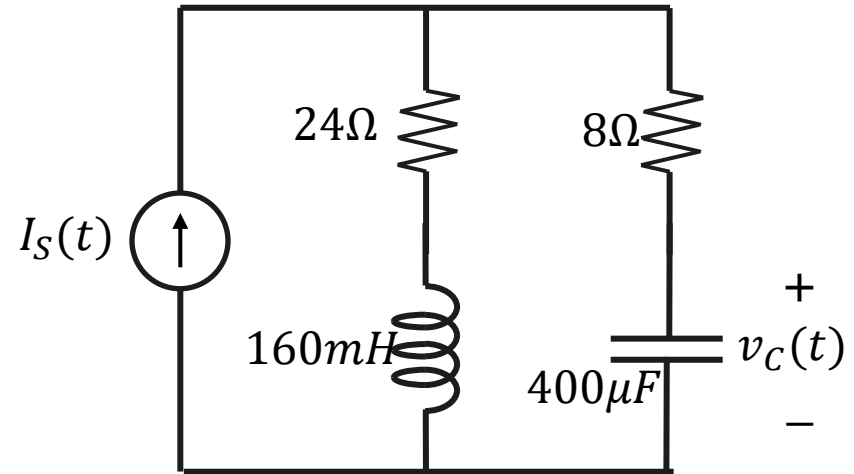
Bonus Material: Example



Example 2.8: In the circuit shown, the current source produces:

$$I_S(t) = \begin{cases} 1/2, & t < 0, \\ 1/4, & t > 0. \end{cases}$$

Find $v_C(t)$ for $t > 0$.



$$\text{Answer: } v_C(t) = \left[6 + 6e^{-100t} \cos(75t) - \frac{1}{3}e^{-100t} \sin(75t) \right] u(t)$$